

**MBA
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UNSUPERVISED MACHINE LEARNING: Simple and Multiple Correspondence Analysis

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Contextualization

- When to apply the correspondence analysis?
 - Adequate technique for the analysis of categorical variables (**qualitative**)
 - The objective is to verify if there is a statistically significant **association** between the variables and their categories, creating the **perceptual map** to visualize the associations
 - If there is a quantitative variable, it is necessary for it goes through a categorization process previously
 - For example: the age is a quantitative variable (25, 42, 73, 81 years) and could be categorized as: 0-30 years is the category 1, 31-60 years is the category 2, 61-90 years is the category 3...

Contextualization

- It is an exploratory technique (not supervised)
 - To evaluate the joint relationship between the variables
 - There are no models of the type “ $y_i = x_{1i} + x_{2i} + \dots + u_i$ ”
 - They are not suitable for inference purposes
 - If new observations are added to the database, it is appropriate to do the analysis again

Contextualization

- Examples of application
 - Income and the credit approval status
 - Educational stage and position in companies
 - Type of soil and culture implemented
 - Severity of the symptoms of the disease and comorbidities
 - Among others...

Contextualization

- Analysis of variables generated by the Likert scale
 - Examples: I strongly agree; I agree; neutral; disagree; strongly disagree;
 - It avoids the problem of arbitrary weighting
 - Each point of the Likert scale becomes a category of the variable in the simple or multiple correspondence analysis

Implementation

Simple Correspondence Analysis

Simple Correspondence Analysis

- Also known as Anacor (Brazilian abbreviation)
 - When the objective is to study **the association between two variables and their categories**
 - It is possible to separate the Anacor (correspondence analysis) into two parts:
 1. Analysis of the statistical significance of the association between the variables and their categories through the chi-square test (χ^2)
 2. Elaboration and interpretation of the perceptual map

1. Analysis of statistical significance (chi-square test)

Simple Correspondence Analysis

1. Contingency table

- It contains the absolute frequencies observed for each pair of categories of the variables
- It is a *cross-tabulation* table

		Variable B					
		Category 1	Category 2	Category 3	...	Category J	Total
Variable A	Category 1	n_{11}	n_{12}	n_{13}	...	n_{1J}	Σ_{L1}
	Category 2	n_{21}	n_{22}	n_{23}	...	n_{2J}	Σ_{L2}
	Category 3	n_{31}	n_{32}	n_{33}	...	n_{3J}	Σ_{L3}
	...	...	...	...	...	...	...
	Category I	n_{I1}	n_{I2}	n_{I3}	...	n_{IJ}	Σ_{LI}
	Total	Σ_{C1}	Σ_{C2}	Σ_{C3}	...	Σ_{CJ}	N

Simple Correspondence Analysis

2. Table of absolute frequencies

- For the cell related to the categories 1 of the two variables, the expected absolute frequency is:

$$\text{Expected absolute frequency} = \frac{(\Sigma_{L1} \times \Sigma_{C1})}{N}$$

- This calculation should be performed for each pair of categories of the variables, changing only the numerator

Simple Correspondence Analysis

3. Residual table

- For the cell related to the categories 1 of the two variables, the residual value is:

$$\text{Residual} = n_{11} - \frac{(\sum_{L1} \times \sum_{C1})}{N}$$

- That is, **residual = absolute frequency observed – expected absolute frequency**
- The same calculation is performed for each pair of categories

Simple Correspondence Analysis

4. Table with the values χ^2

- For the cell related to the categories 1 of the two variables, the value of the statistics χ^2 is:

$$\chi^2 = \frac{(\text{residual}_{11})^2}{(\text{absolute expected frequency}_{11})}$$

- The same calculation is performed for each pair of categories, and then the values of all cells are added

Simple Correspondence Analysis

4. Table with the values χ^2

- The objective is to verify if there is a statistically significant association between the variables (using the sum of the χ^2)
 - H_0 : the variables are associated in a random way
 - H_1 : the association between the variables is not done in a random way
- Considering the level of significance and the degrees of freedom, if the value of χ^2 statistics is greater than its critical value, there is a significant association between the two variables (H_1)
 - Degrees of freedom = $(I - 1) \times (J - 1)$

Simple Correspondence Analysis

5. Table of standardized (and adjusted standardized) residuals

- While the analysis of the χ^2 allows us to verify if there is the dependence between the two variables, the analysis of adjusted standardized residuals allows deepening the analysis focusing on the categories of the variables
- How do the categories of a variable relate to the categories of the other variable?
 - Therefore, the excess or lack of occurrence of cases in the categories of the two variables is observed.

Simple Correspondence Analysis

5. Table of standardized residuals

- For the cell related to the categories 1 of the two variables, the standardized residual value is:

$$\text{Standardized Residual} = \frac{\text{residual}_{11}}{\sqrt{(\text{absolute expected frequency}_{11})}}$$

- The same calculation is performed for each pair of categories

Simple Correspondence Analysis

5. Table of adjusted standardized residuals

- For the cell related to the categories 1 of the two variables, the adjusted standardized residual value is:

$$\text{Adjusted standardized Residual} = \frac{\text{standard residual}_{11}}{\sqrt{\left(1 - \frac{\sum c_1}{N}\right) \times \left(1 - \frac{\sum l_1}{N}\right)}}$$

- The same calculation is performed for each pair of categories

Simple Correspondence Analysis

5. Table of adjusted standardized residuals

- If the adjusted standardized residual value in a cell is greater than **1.96**, the interpretation is that there is a significant association at the significance level of **5%** between the two categories that interact in the cell; if it is less than 1.96, there is no statistically significant association
 - The reference of 1.96 is the critical value of the standard normal for the significance level of 5%

2. Elaboration and interpretation of the perceptual map

Simple Correspondence Analysis

1. Determine the eigenvalues (λ^2)

- The quantity (m) of eigenvalues depends on the quantity of categories in the variables:
 $m = \min(I - 1, J - 1)$
- In the Anacor (correspondence analysis), the eigenvalues refer to the partial principal inertia and are the basis for determining the total main inertia and the percentage of the total principal inertia in each dimension of the perceptual map

Simple Correspondence Analysis

1. Determine the eigenvalues (λ^2)

- As a basis for the calculation of the eigenvalues, a matrix A is defined
 - A way of obtaining the matrix A, based on the previous steps, is to make for each cell of the standardized residuals matrix the following calculation:

(*Standardized residual*)

$\sqrt{N}$

- Based on the matrix A, the matrix W: **$W = A'$** is obtained. A

Simple Correspondence Analysis

1. Determine the eigenvalues (λ^2)

- Identifying **W**, the eigenvalues are obtained by the solution of the following expression: $\det(\lambda^2 \cdot I - W) = 0$

$$\begin{vmatrix} \lambda^2 - w_{11} & -w_{12} & -w_{13} \\ -w_{21} & \lambda^2 - w_{22} & -w_{23} \\ -w_{31} & -w_{32} & \lambda^2 - w_{33} \end{vmatrix} = 0$$

- I** is the identity matrix

Simple Correspondence Analysis

1. Determine the eigenvalues (λ^2)

- Based on the eigenvalues (λ^2), the percentage of the total principal inertia of each dimension

is found

$$\% \text{ of the Total Principal Inertia (for each dimension)} = \frac{\lambda^2_{\text{dimension}}}{\lambda^2_{\text{total}}}$$

- The greater the total principal inertia (and the χ^2), the stronger the association is between the variables in analysis

$$\text{Total Principal Inertia} = \frac{\chi^2}{N}$$

Simple Correspondence Analysis

2. Determine the row and column masses

- The **masses** represent the influence that each category exercises on the other *categories* of their variable, whether in the column or *row profiles*.
- Based on the “totals” of the contingency table, the average masses are obtained for the category 1 of the variables:

$$\text{Mass in the column} = \frac{\Sigma_{L1}}{N} \qquad \text{Mass in the row} = \frac{\Sigma_{C1}}{N}$$

- The same calculation is performed for the other categories

Simple Correspondence Analysis

3. Determine the eigenvalues

- For the **W** matrix, it is possible to find the eigenvectors from the eigenvalues (λ^2) calculated
- Replacing eigenvalues of each dimension in the matrix defined as **$\det(\lambda^2 \cdot \mathbf{I} - \mathbf{W}) = 0$** and solving the equations system that originate from it, it is possible to find the eigenvectors of the column (V), and based on them, to find the eigenvectors of the row (U)

$$u'_k = \underbrace{[D_i^{-1/2} \cdot (P - Ic') \cdot Dc^{-1/2}]}_{\text{It is the matrix A}} \cdot v'_k \cdot \lambda_k^{-1}$$

It is the matrix A

Simple Correspondence Analysis

4. Determine the coordinates of the categories

- Variable **in row** in the contingency table

- Coordinates of abscissas (X)

$$X_i = \sqrt{\lambda_1} \cdot D_i^{-1/2} \cdot u_{i1}$$

- Coordinates of ordinates (Y)

$$Y_i = \sqrt{\lambda_2} \cdot D_i^{-1/2} \cdot u_{i2}$$

- Coordinates of the K-th dimension (k = quantity of λ)

$$Z_i = \sqrt{\lambda_k} \cdot D_i^{-1/2} \cdot u_{ik}$$

Simple Correspondence Analysis

4. Determine the coordinates of the categories

- Variable **in column** in the contingency table

- Coordinates of abscissas (X)

$$X_c = \sqrt{\lambda_1} \cdot Dc^{-1/2} \cdot v_1$$

- Coordinates of ordinates (Y)

$$Y_c = \sqrt{\lambda_2} \cdot Dc^{-1/2} \cdot v_2$$

- Coordinates of the K-th dimension (k = quantity of λ)

$$Z_c = \sqrt{\lambda_k} \cdot Dc^{-1/2} \cdot v_k$$

Multiple Correspondence Analysis (MCA)

Multiple Correspondence Analysis

- Association between more than two variables
 - Only variables that present a statistically significant association with at least another variable contained in the analysis participate in the MCA
 - Before elaborating the MCA, it is important to perform a χ^2 test for each pair of variables
 - If any of them do not present association with others, it is not included in the correspondence analysis

Multiple Correspondence Analysis

1st method: using **the Binary Matrix**

- The binary matrix is obtained by the transformation of the qualitative variables into binary variables, that is, values 0 or 1
- Based on the binary matrix (Z), the total principal inertia can be obtained in the MCA
- Assuming that the binary matrix (Z) is similar to a contingency table of the Anacor (correspondence analysis), it is possible to obtain the partial principal inertia of the dimensions, eigenvalues, eigenvectors and coordinates of this matrix

Multiple Correspondence Analysis

1st method: using the Binary Matrix

- An example of a binary matrix (Z) is:

P.S.:	Variable A		Variable B			Variable C			
	Category 1	Category 2	Category 1	Category 2	Category 3	Category 1	Category 2	Category 3	Category 4
1	1	0	0	1	0	1	0	0	0
2	0	1	0	0	1	0	1	0	0
3	0	1	1	0	0	0	0	1	0
N	1	0	1	0	0	0	0	0	1

- Quantity of dimensions (λ^2) = $J - Q = (9 - 3) = 6$, in which “J” is the total quantity of categories and “Q” the quantity of variables

Multiple Correspondence Analysis

2nd method: using **the Burt Matrix**

- The Burt matrix is defined as: $\mathbf{B} = \mathbf{Z}' \cdot \mathbf{Z}$
 - It is possible to combine the contingency tables with the crossing of all variable pairs in a single matrix
 - When considering the Burt matrix as a contingency table, it is possible to perform an Anacor and obtain the coordinates of the categories

Reference

Fávero, Luiz Paulo; Belfiore, Patrícia. (2017).

Manual de análise de dados: estatística e modelagem multivariada com Excel®, SPSS® e Stata®. Rio de Janeiro: Elsevier